JAKARIA HABIB

Department of Information and Communication Engineering (ICE) · Pabna University of Science and Technology (PUST), Bangladesh

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Research Interests

Physics-Informed AI & Medical Perception: Biomechanical Neuromuscular Simulation, Vision Transformers in Healthcare & Remote Sensing.

Foundation Models & Learning Dynamics: Continual Reinforcement Learning, Faithfulness in Reasoning, Geometric Deep Learning.

Embodied AI & Robotics: Action-Conditioned World Models, Safe Robot Perception, Path Planning, Digital Twins.

Education

Bachelor of Science (Engineering) in Information and Communication Engineering **2022 – 2026**
Pabna University of Science and Technology (PUST), Pabna, Bangladesh **CGPA: 3.08 / 4.00**

- **Undergraduate Research Thesis:** *"Physics-Based Inverse Modeling for Sub-Millisecond Neural Delay Estimation in Human Locomotion"* (Advisor: Md. Imran Hossain, Faculty, Dept. of ICE, PUST).
  Formulated an end-to-end OpenSim pipeline inverting Hill-type activation ODEs across **45,445** synthetic gait trials, estimating latent delays with **MAE = 0.10 ms** and $R^2 = 0.998$. Validated that **93.6%** of timing information resides in muscle activation versus 6.4% in kinematics; verified structural stability under +7% femoral mass perturbation.
- **Curriculum Rigor & Advanced Thematic Coursework** *(Mapped from Official ICE Degree Syllabus)*:
  ◦ *Math & Statistics:* Linear Algebra (Math-2101), Complex Variables & Transforms (Math-2201), Calculus, Differential Equations, Statistics, Probability & Hypothesis Testing.
  ◦ *AI, Perception & Signals:* Neural Networks (ICE-4205), Computer Vision (ICE-4214), AI & Robotics (ICE-3101), Digital Image Processing (ICE-3207), DSP (ICE-3109), Signals & Systems (ICE-2205).
  ◦ *Algorithms & Systems Architecture:* Data Structures & Algorithms (ICE-2201), Discrete Math (ICE-2105), OOP (ICE-2103), Computer Architecture & Microcontrollers.

Research Publications

(indicates equal contribution or core student researcher; bold indicates self; citations, DOIs and preprints verified)*

Peer-Reviewed SCI-Indexed Journal Articles

1. **Jakaria Habib**, Imran Hossain, Md. Basim Al Zabir Shammo, Taz Warul Mulk, Estiyak Rubaiat. (2026). **"Vision Transformer-Based Damage Assessment from Post-Disaster Satellite Imagery: An Applied Study on Hurricane Harvey."** *Earth Science Informatics*, Springer Nature. **[Springer Nature | SCI-Expanded | Scopus Q1 (SJR: 0.761) | Impact Factor: 4.2]** (First Author)

Top-Tier (A*) Conference Manuscripts

2. Noor Islam S. Mohammad, **Jakaria Habib**, Md. Basim Al Zabir Shammo. (2026). **"Specialist Drift: Continual RL for Capability-Preserving Foundation Model Post-Training."** *Under Review at International Conference on Learning Representations (ICLR 2027).* OpenReview Submission ID: Specialist-Drift-Continual-RL.
3. Noor Islam S. Mohammad, Md. Basim Al Zabir Shammo, Hasan Siddiki, Mahmudul Hasan, Md. Faisal Sheikh, **Jakaria Habib**. (2026). **"SIFT: Robust Meta-Faithfulness Verification of Chain-of-Thought Reasoning Under Distribution Shift."** *Under Review at International Conference on Learning Representations (ICLR 2027).* OpenReview Submission ID: SIFT-Verification.

Peer-Reviewed Workshop Proceedings at Premier International Venues (NeurIPS 2026)

4. **Jakaria Habib**, et al. (2026). **"The Symmetry Budget: How Much Group Structure Should a Network Architecture Assume?"** ***NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations (NeurReps)***. **[NeurIPS 2026 | Accepted Proceedings Track | To Appear in PMLR]**
5. **Jakaria Habib**, et al. (2026). **"HALT: Certifying Trust Horizons in Action-Conditioned Robotic World Models."** ***NeurIPS 2026 Robotics World Modeling Workshop (RWM)***. **[NeurIPS 2026 | Accepted Workshop Poster]**

Peer-Reviewed International Conference Proceedings (Published & Indexed)

6. **Jakaria Habib**, et al. (2026). **"A Hybrid CNN–Transformer Framework for Automated Poultry Disease Classification Using Fecal Images."** *In Proceedings of the 2026 IEEE International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII 2026)*, IEEE Xplore. **[Published & Indexed in IEEE Xplore]** (First Author)
7. Md. Rashed, Md. Basim Al Zabir Shammo, Md. Imran Hossain, Md. Anwar Hossain, Lamia Islam, **Jakaria Habib**. (2025). **"A Machine Learning Framework for Stress Detection Using Photoplethysmography (PPG) Signals."** *In Proceedings of the 2025 IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2025)*, IEEE Xplore. Paper ID: 1435. **[Published & Indexed in IEEE Xplore]**
8. **Jakaria Habib**, et al. (2025). **"The Optimization of Violence Detection and Classification with Vision Transformer and ResNet18 Hybrid Model."** *In Proceedings of the 11th IEEE Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE 2025)*, IEEE Xplore. **[Published & Indexed in IEEE Xplore]**

Journal Manuscripts Under Active Peer Review

9. **Jakaria Habib**, et al. (2026). **"GeoMind: An Autonomous Multi-Agent Framework for Verified Geospatial Analysis."** *Earth Science Informatics*, Springer Nature *(First Author)*.
10. **Jakaria Habib**, et al. (2026). **"The Limits of Physics-Informed Digital Twins Under Transient Loads: An HIL Study for Micro-Mechatronics."** *Journal of Intelligent Manufacturing*, Springer Nature.

Learned Societies, Scholarly Credentials & Presentations

- **American Physical Society (APS), USA:** Active Professional Member (Account ID: `62251156`, Term: Sept 2026 – Aug 2027).
- **Nature Masterclasses (Springer Nature):** Certified in *"Focus on Peer Review"* (Signed by Dr. Magdalena Skipper, Editor-in-Chief, *Nature*), Issued May 2026. Rigorous training in scientific critique, peer-review methodology, and publication ethics.
- **Conference Oral Speaker, IEEE QPAIN 2025:** Delivered official conference presentation for Paper ID 1435: *"A Machine Learning Framework for Stress Detection Using Photoplethysmography (PPG) Signals"*, BAUST, July 31 – Aug 2, 2025.
- **Student Member & Volunteer, IEEE PUST Student Branch (2024 – Present):** Organized and conducted student technical workshops on machine learning workflows and scientific computing with Python.

Selected Research & Robotics Projects

- **Vision-Guided Autonomous Inspection Mobile Robot** *Embedded Hardware & Vision*
  Engineered a custom 3D-printed chassis robot with Ackermann steering servos and camera perception; integrated deep learning crack segmentation with path coverage planning algorithms for automated structural inspection.
- **OpenSim Biomechanical Neuromuscular Latency Estimator** *Python / OpenSim API*
  Interfaced Python with OpenSim musculoskeletal physics simulation engine to extract high-dimensional joint torques and activations; implemented SHAP and gradient feature importance for interpretable latent delay modeling.

Verified Academic Benchmarks & Certifications

- **HarvardX (Harvard University):** *CS109x: Introduction to Data Science with Python* (Course Director: Dr. Pavlos Protopapas, Scientific Program Director, Harvard), Issued Feb 2026. Verified ID: `61d90c194b97494ca85bc68f25086b53`.
- **Columbia University (Columbia+):** *Learning AI Through Visualization*, Issued Aug 2025. Verified Performance: **83.00 / 84.00 (98.8% Grade)**. Credential ID: `158822783`.

Technical Skills & Tooling

- **Programming Languages:** Python, MATLAB, C/C++, Bash, SQL.
- **Deep Learning & Perception:** PyTorch, Hugging Face Transformers, TensorFlow, Scikit-learn, OpenCV, Torchvision.
- **Physics, Robotics & Simulation:** OpenSim (Musculoskeletal Modeling), ROS (Robot Operating System), Gazebo.
- **Scientific Workflow & Languages:** Git, Linux/Unix, LaTeX, Weights & Biases, VS Code | Bengali (Native), English (Fluent).